CHEMICAL AUTHORIZATION SYSTEM

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the requirements for the award of the degree of

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From

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Submitted By

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CERTIFICATE

*This is to certify that the Project report entitled "Central Service Portal" is a bonafide record of the work done by **MUHAMMED RINSHAD (MEA25MCA-2014)** under our supervision and guidance. The report has been submitted in fulfillment of the requirement for award of the Degree of Master of Computer Applications from the APJ Abdul Kalam Kerala Technological University for the year 2026.*

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ABSTRACT

The purchase of chemicals by laboratories and individuals often involves manual verification procedures, paper-based documentation, and offline communication, making the overall process time-consuming and difficult to manage. Existing systems lack a centralized platform to verify purchase authorizations, monitor laboratory inventories, and track purchase requests efficiently. As a result, users frequently experience delays, laboratories struggle with stock management, and administrators face difficulties in maintaining transparency and preventing unauthorized chemical purchases.

To overcome these challenges, I developed a **Chemical Purchase Authorization and Laboratory Management System**, a secure web-based application that digitizes the complete chemical purchasing workflow. The system enables users to apply for a Chemical Purchase Authorization, browse licensed laboratories, view available chemicals, and submit purchase requests using a valid Authorization ID. Laboratory staff can verify authorizations, check stock availability, approve or reject requests, reserve chemicals for offline purchases, and process online purchase requests efficiently. Administrators are responsible for managing licensed laboratories, approving authorization applications, monitoring users, and supervising the overall operation of the system.

The application provides a centralized platform that improves transparency, reduces manual paperwork, minimizes processing delays, and enhances inventory management. It also enables users to track authorization status, purchase requests, reservations, notifications, and order updates through a user-friendly interface. The system was developed using **HTML, CSS, and JavaScript** for the frontend, **Node.js with Express.js** for the backend, and **MySQL** as the database to provide secure, reliable, and efficient management of chemical purchase authorization and laboratory operations

CONTENTS

1. INTRODUCTION **1**

2. SYSTEM ANALYSIS **3**

2.1 Existing System 4

2.2 Proposed System 4

2.3 Module Description 4

2.4 Sprint 5

3. FEASIBILITY STUDY **8**

3.1 Economical Feasibility 9

3.2 Technical Feasibility 9

3.3 Operational Feasibility 9

3.4 Behavioural Feasibility 9

3.5 Software Feasibility 10

3.6 Hardware Feasibility 10

4. SOFTWARE ENGINEERING PARADIGM **11**

4.1 Agile Model 12

4.2 Scrum 12

5. SYSTEM REQUIREMENT SPECIFICATION **13**

5.1 Software Requirements 14

5.2 Hardware Requirements 14

6. SYSTEM DESIGN **15**

6.1 Database Design 16

6.2 Tables 18

6.3 UML Design 20

6.4 Use Case Diagram 21

6.5 Scenario 22

6.6 Data flow diagram 23

7. SYSTEM DEVELOPMENT **27**

7.1 Coding 28

8. SYSTEM TESTING AND IMPLEMENTATION **31**

8.1 Types of Testing 32

8.2 Implementation 34

9. SYSTEM MAINTENANCE **35**

10. FUTURE ENHANCEMENT **37**

11. CONCLUSION **39**

12. APPENDIX **41**

13. BIBLIOGRAPHY **48**

INTRODUCTION

1. INTRODUCTION

The safe distribution and purchase of chemicals require proper authorization, inventory management, and continuous monitoring to prevent unauthorized access and ensure regulatory compliance. However, many laboratories still depend on traditional manual procedures for handling chemical purchase requests, verifying customer authorizations, and maintaining inventory records. These methods are often slow, error-prone, and inefficient, resulting in delays, inconsistent stock records, and limited transparency throughout the purchasing process.

To address these issues, I developed a **Chemical Purchase Authorization and Laboratory Management System**, a secure web-based platform designed to automate and simplify the complete chemical purchasing workflow. The system provides a centralized environment where users can register, apply for Chemical Purchase Authorization by submitting the required documents, and receive a unique Authorization ID after approval. Using this authorization, users can browse licensed laboratories, check chemical availability, and submit purchase requests conveniently through the application.

The platform also enables laboratory staff to verify Authorization IDs, review purchase requests, monitor chemical inventory, reserve stock for offline purchases, and process online purchase requests efficiently. Administrators are responsible for registering licensed laboratories, approving authorization applications, managing users, and monitoring overall system activities. By integrating these operations into a single platform, the system improves transparency, accountability, and operational efficiency while significantly reducing manual effort.

SYSTEM ANALYSIS

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2. SYSTEM ANALYSIS

2.1 Existing System

In the existing system, chemical purchase requests are handled manually using paper records and offline verification. Users must visit laboratories to submit documents and confirm chemical availability. Laboratory staff manually verify authorization details and maintain inventory records, making the process slow, inefficient, and prone to errors. The absence of a centralized system also makes it difficult to monitor stock, purchase requests, and user authorizations effectively.

2.2 Proposed System

The proposed **Chemical Purchase Authorization and Laboratory Management System** provides a centralized web-based platform for managing the complete chemical purchasing process. Users can apply for authorization, browse licensed laboratories, view available chemicals, and submit purchase requests online. Laboratory staff verify Authorization IDs, manage inventory, approve or reject requests, and reserve stock when necessary. The system improves efficiency, transparency, inventory management, and request tracking.

2.3 Module description

The system is divided into 3 modules based on user roles, each with specific functionalities:

Admin

- View system dashboard and statistics
- Register and manage licensed laboratories
- Manage users
- Approve or reject Chemical Purchase Authorization applications
- View all registered users and laboratories

Laboratory

- Register laboratory after license approval
- Manage laboratory profile
- View and process purchase requests
- Verify Authorization ID
- Approve or reject purchase requests
- Reserve stock for offline purchases

User

- Register and login
- Apply for Chemical Purchase Authorization
- View Authorization ID and validity
- Browse licensed laboratories
- View available chemicals and stock
- Submit chemical purchase requests

2.4 Sprint

Sprint 1

Module	Task	Pending task if any	Hour of completion	Expected date of completion	Actual date of completion	Reason for delay
User	Login page	-	2 hr.	15/07/2026	15/07/2026	-
	Authorization page	-	3 hr.	17/07/2026	17/07/2026	-
	Laboratories page	-	2 hr.	18/07/2026	18/07/2026	-
	Purchase page	-	3 hr.	21/07/2026	21/07/2026	-
	Profile page	-	2 hr.	22/07/2026	22/07/2026	-

Sprint 2

Module	Task	Pending task if any	Hour of completion	Expected date of completion	Actual date of completion	Reason for delay
Laboratory	Login Page	-	1 hr.	30/08/2026	30/08/2026	-
	Profile page	-	2 hr.	01/08/2026	01/08/2026	-
	Chemical management Page	-	3 hr.	05/08/2026	05/08/2026	-
	Inventory Dashboard Page	-	1 hr.	07/08/2026	07/08/2026	-
	Purchase request page	-	2 hr.	15/08/2026	15/08/2026	-
	Purchase reports page	-	1 hr.	17/08/2026	17/08/2026	-

Sprint 3

Module	Task	Pending task if any	Hour of completion	Expected date of completion	Actual date of completion	Reason for delay
Admin		-				-
		-				-
		-				-
		-				-
		-				-
		-				-
		-				-
		-				-
		-				-
		-				-
		-				-

FEASIBILITY STUDY

3.FEASIBILITY STUDY

An analysis of the ability to complete a project successfully, taking into account legal, economic, technological, scheduling, and other factors is considered as feasibility study. Rather than just diving into a project and hoping for the best, feasibility study allows project managers to investigate the possible negative and positive outcomes of a project before investing too much time and money.

3.1 Economical Feasibility

The project is economically feasible because it is developed using open-source technologies, which significantly reduce software development costs. The system also minimizes manual paperwork, improves inventory management, and reduces the time required for processing chemical purchase requests, making it a cost-effective solution for laboratories

3.2Technical Feasibility

The proposed system is technically feasible because it is developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. These technologies are reliable, secure, and suitable for developing a web-based application. The modular architecture makes the system easy to maintain, update, and expand with additional features in the future.

3.3Operational Feasibility

The proposed system provides a simple and user-friendly interface for users, laboratory staff, and administrators. It automates the chemical purchase authorization process, inventory management, and request tracking, improving operational efficiency while reducing manual work and processing delays.

3.4 Behavioral Feasibility

The system is designed to be easy to learn and use, requiring minimal training for all users. Clear navigation, simple forms, and real-time status updates encourage users to adopt the system quickly. The digital workflow also improves communication between users, laboratories, and administrators.

3.5 Software Feasibility

The application is developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. It can be accessed through modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox. The system supports Windows, Linux, and macOS without requiring additional software installation.

3.6 Hardware Feasibility

The proposed system requires only a standard desktop, laptop, or smartphone with an internet connection and a modern web browser. Laboratories and administrators can operate the system using existing computer systems, eliminating the need for specialized hardware and reducing implementation costs.

SOFTWARE ENGINEERING PARADIGM

4. SOFTWARE ENGINEERING PARADIGM

The software engineering paradigm which is also referred to as a software process model or Software Development Life Cycle (SDLC) model is the development strategy that encompasses the process, methods and tools. SDLC describes the period of time that starts with the software system being conceptualized.

4.1 Agile model

Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product. Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. At the end of the iteration, a working product is displayed to the customer and important stakeholders.

In Agile, the tasks are divided to time boxes (small time frames) to deliver specific features for a release. Iterative approach is taken and working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer.

4.2 Scrum

Scrum is an agile framework for managing knowledge work, with an emphasis on software development. It is designed for teams of three to nine members, who break their work into actions that can be completed within time boxed iterations, called "sprints", no longer than one month and most commonly two weeks, then track progress and re-plan in 15-minute stand-up meetings, called daily scrums.

Scrum is an iterative and incremental framework for product development. It promotes a flexible strategy where development teams collaborate to achieve common goals, challenging traditional sequential methods. Scrum encourages self-organization, physical co-location, and daily communication among team members.

SYSTEM REQUIREMENTS SPECIFICATION

5. SYSTEM REQUIREMENTS SPECIFICATION

5.1 Software Requirements

Component	**Requirement**
Operating system	Windows 8 or above
Backend	Node.js
Frontend	Html,CSS,Javascript,Bootstrap
Database	Mysql
IDE	Vs code

5.2 Hardware Requirements

Component	**Requirement**
Processor	Intel core i3 or above
RAM	Minimum 8GB (recommended 16GB)
Monitor	14" LCD or Above
Key Board	108 Keys
Mouse	Any type of mouse

SYSTEM DESIGN

6. SYSTEM DESIGN

System design is the first in the development phase for many engineered product or system. It may define the process of applying various techniques and principles for the purpose of defining a device, a process or system in sufficient detail to permit its physical realization.

6.1 Database Design

Database design is the process of producing a detailed data model of a database. This logical data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a data definition language, which can then be used to create a database.

In the relational model, tables and views represent data, while in an object database, entities and relationships correspond to object classes. Database design encompasses not only base data structures but also forms and queries within a database management system. The design process involves several steps, primarily focusing on determining relationships between data elements and establishing a logical structure, leading to normalization.

Normalization

It is the procedure of transforming a relationship into a standard form. This process addresses issues that may arise from data redundancy, which refers to the repetition of data within the database. Additionally, it helps maintain data integrity and resolves potential problems related to insertion, updating, and deletion anomalies.

Decomposing is the process of breaking down relations into several smaller relations to eliminate anomalies and uphold data integrity. To achieve this, normal forms or guidelines for structuring relations are utilized.

Insertion anomaly: Inability to add data to the database due to absence of other data.

Deletion anomaly: Unintended loss of data due to deletion of other data.

Update anomaly: Data inconsistency resulting from data redundancy and partial update.

Normal Forms: These are the rules for structuring relations that eliminate anomalies.

1. First Normal Form (1NF)

A relation is considered to be in first normal form when all values within that relation are atomic for each attribute. This means that no attribute value can consist of a set of values or, as it's often described, a repeating group.

2. Second Normal Form (2NF)

A relation is in second normal form if it is in first normal form and meets at least one of the following rules.

- Primary key is not a composite primary key
- No non key attributes are present
- Every non key attribute is fully functionally dependent on full set of primary keys.

3. Third normal Form (3NF)

A relation is said to be in third normal form if there exist no transitive dependencies.

Transitive dependency occurs when two non-key attributes depend on each other and on the primary key. This normalization principle helps decompose data into multiple tables, ensuring consistent data maintenance.

In Third Normal Form (3NF), the main goal is to remove transitive dependencies, which means that non-key attributes should depend only on the primary key, not on other non-key attributes. This ensures that every piece of data in a table describes only the entity represented by that table. By eliminating transitive dependencies, we avoid redundancy and make updates, insertions, and deletions more efficient and error-free.

Overall, 3NF improves database design by making it cleaner, more flexible, and easier to maintain. It minimizes duplicate data and maintains data consistency across relationships, which is crucial for large and frequently updated databases.

6.2 Tables

User

FIELD NAME	DATA TYPE	CONSTRAINTS
User_id	INT	Primary Key
Full_name	VARCHAR	NOT NULL
email	VARCHAR	UNIQUE,NOT NULL
password	VARCHAR	NOT NULL
address	VARCHAR	NOT NULL
phone	VARCHAR	NOT NULL
status	ENUM('active','inactive')	NOT NULL

Chemicals

FIELD NAME	DATA TYPE	CONSTRAINTS
chemical_id	INT	Primary Key
lab_id	INT	Foreign Key
Chemical_code	VARCHAR	NOT NULL
Chemical_name	VARCHAR	NOT NULL
formula	VARCHAR	NOT NULL
unit	ENUM('liter','gram')	NOT NULL
price	DECIMAL(10,2)	NOT NULL
stock	DECIMAL(10,2)	NULL
reserved_stock	DECIMAL(10,2)	NULL
Created_at	TIMESTAMP	DEFAULT_GENERATED

Laboratories

FIELD NAME	DATA TYPE	CONSTRAINTS
lab_id	INT	Primary Key
license_id	INT	Foreign Key
lab_name	VARCHAR	NOT NULL
owner_name	VARCHAR	NOT NULL
email	VARCHAR	UNIQUE,NOT NULL
password	VARCHAR	NOT NULL
phone	VARCHAR	NOT NULL
address	TEXT	NOT NULL

Purchase_requests

FIELD NAME	DATA TYPE	CONSTRAINTS
request_id	INT	Primary Key
user_id	INT	Foreign Key
lab_id	INT	Foreign Key
quantity	DECMIAL(10,2)	NOT NULL
purchase_mode	ENUM('offline','online')	NOT NULL
request_status	ENUM('submited','aproved')	NOT NULL
request_date	TIMESTAMP	NULL
purchase_code	VARCHAR	NULL
reservation_start	DATETIME	NULL
reservation_expiry	DATETIME	NULL

6.3 UML Designs

The unified Modelling Language (UML) is a standard language for specifying, visualizing, constructing, and documenting the artifacts of the software systems, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing object-oriented software and software development process. The UML uses mostly graphical notations to express the design of software projects. Using the UML helps project teams communicate, explore potential designs, and validate the architectural design of the software.

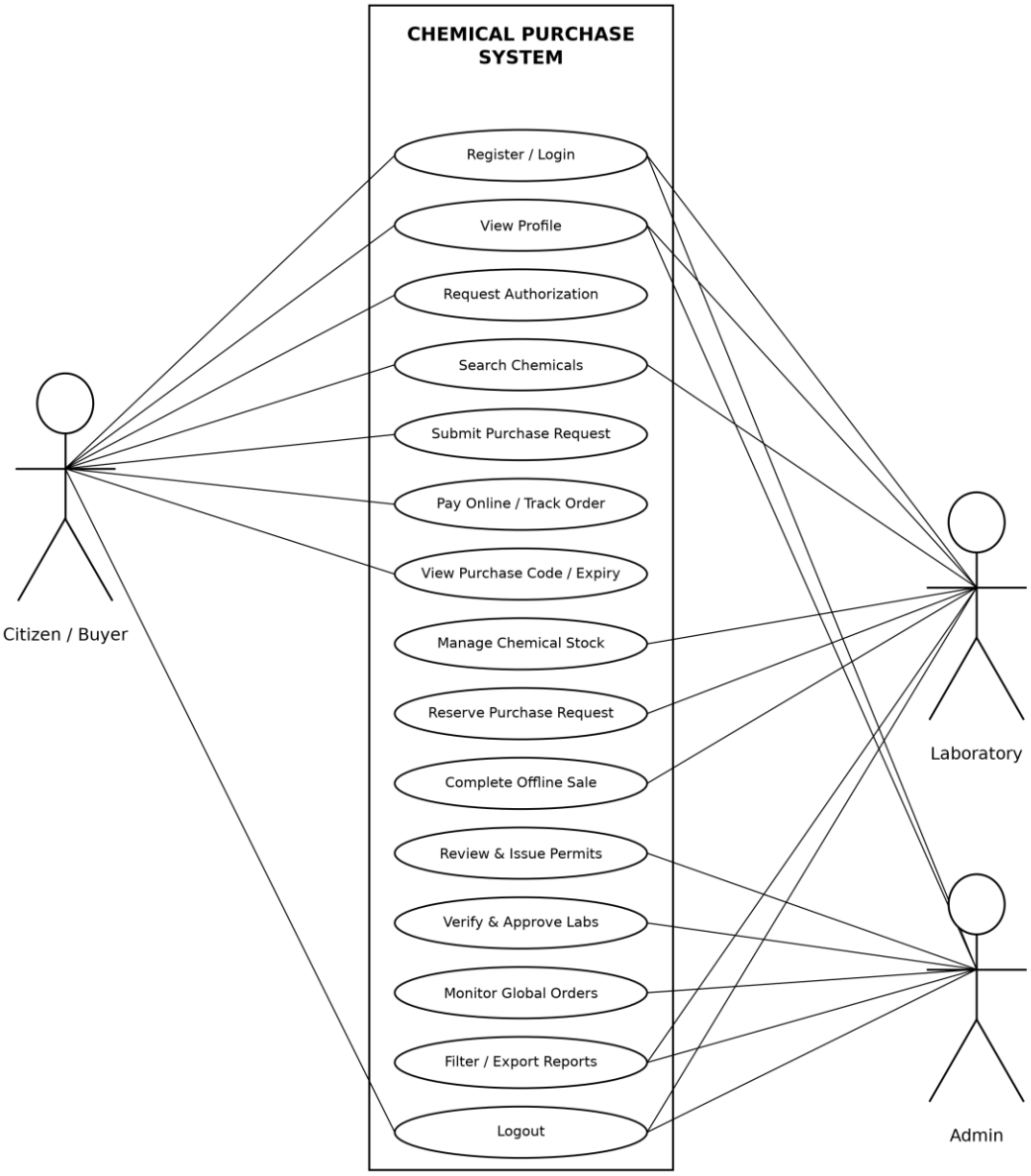
6.4 User Story

As a researcher or science student, I want to register on the Chemical Purchase System, submit my institutional identification and research proposal for regulatory verification, receive an approved authorization code, search certified laboratory suppliers for chemicals by name, formula, or chemical code, and request the exact quantity required for my research so that I can legally and safely obtain chemicals. I want to choose between online purchase and offline pickup, where online orders can be paid through an integrated payment gateway, tracked through preparation and transit, and automatically completed after the defined fulfillment period, while offline orders generate a unique verification code and a 48-hour reservation period for collection at the laboratory counter. As a laboratory manager, I want to register using a verified laboratory license, manage chemical catalogs and inventory, search stock, review authorized purchase requests, reserve requested quantities, verify offline pickup codes, fulfill online orders, and export order records for accounting and compliance purposes. As a regulatory administrator, I want to maintain the laboratory license registry, review researcher authorization applications, issue secure authorization codes with expiry dates, monitor chemical transactions, and maintain an auditable record of chemical movements. As the system automation engine, I want to automatically reserve requested stock, finalize eligible online orders, expire uncollected offline reservations after 48 hours, invalidate expired verification codes, restore released quantities to available inventory, and maintain accurate buyer, laboratory, and transaction histories so that inventory and regulatory records remain synchronized.

6.5 Use case diagram

A use case diagram is a visual representation that shows how different users (called actors) interact with a system to achieve specific goals. It's one of the primary diagrams used in Unified Modeling Language (UML) to capture functional requirements. Each use case represents a particular function or service the system provides, and the relationships between actors and use cases show who can perform which actions. This helps developers and stakeholders quickly understand what the system is supposed to do from a user's perspective, without needing to go deep into technical details.

In software development, use case diagrams are especially useful during the analysis and design phase. They help identify the main system functions, clarify user roles, and ensure that all requirements are captured before development begins. These diagrams serve as a communication bridge between clients, developers, and testers, reducing misunderstandings. By visually outlining the interactions, they make it easier to plan, document, and verify system behavior in real-world scenarios.



Use Case Diagram – Chemical Purchase System

6.5 Scenario

A scenario is a detailed description of a particular situation or sequence of events that helps illustrate how a system or process works. In software engineering, scenarios are often used to show how users interact with a system to achieve specific goals. For example, a scenario might describe how a student logs into a college portal, applies for leave, and checks approval status. Scenarios make it easier to understand user needs, test different conditions, and visualize the flow of operations within a system. They serve as practical examples that bridge the gap between theoretical requirements and real-world usage.

BUYER / USER

- Citizen can Register and Login.
- Citizen can View and Edit Profile Information.
- Citizen can Request Chemical Authorization with Document Upload.
- Citizen can Track Status of Authorization Requests.
- Citizen can View Approved Authorization Codes and Expiry Dates.
- Citizen can Search and Browse Chemicals by Name, Code, or Formula.
- Citizen can Submit Purchase Requests (Online or Offline Mode).
- Citizen can Pay Online and Track Delivery Milestones in Real Time.
- Citizen can View Offline Purchase Code and Reservation Countdown Timer.
- Citizen can View Dashboard Statistics and Completed Purchase History.
- Citizen can Logout.

LABORATORY

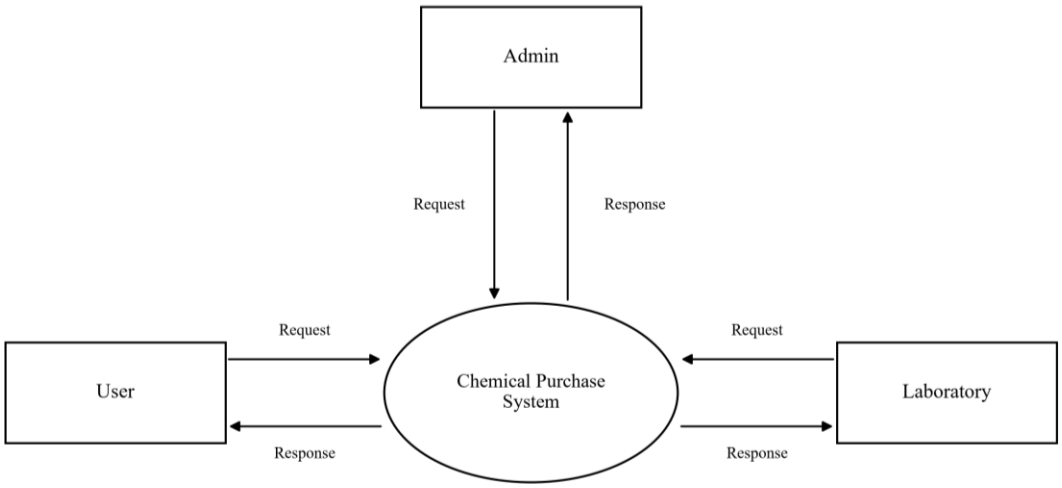
- Laboratory can Register with License ID and Login after Approval.
- Laboratory can View and Update Laboratory Profile Details.
- Laboratory can Add and Manage Chemicals with Formula, Stock, and Price.
- Laboratory can Search Inventory by Chemical Formula, Name, or Code.
- Laboratory can Filter Chemicals by Category and Availability Status.
- Laboratory can Review and Approve Buyer Purchase Requests.
- Laboratory can Reserve Stock and Issue Unique Purchase Codes.
- Laboratory can Verify Purchase Codes and Complete Offline Counter Sales.
- Laboratory can Expire Unpaid Reservations to Automatically Restore Stock.
- Laboratory can View Inventory Dashboard and Export Reports to Excel.
- Laboratory can Logout.

ADMIN

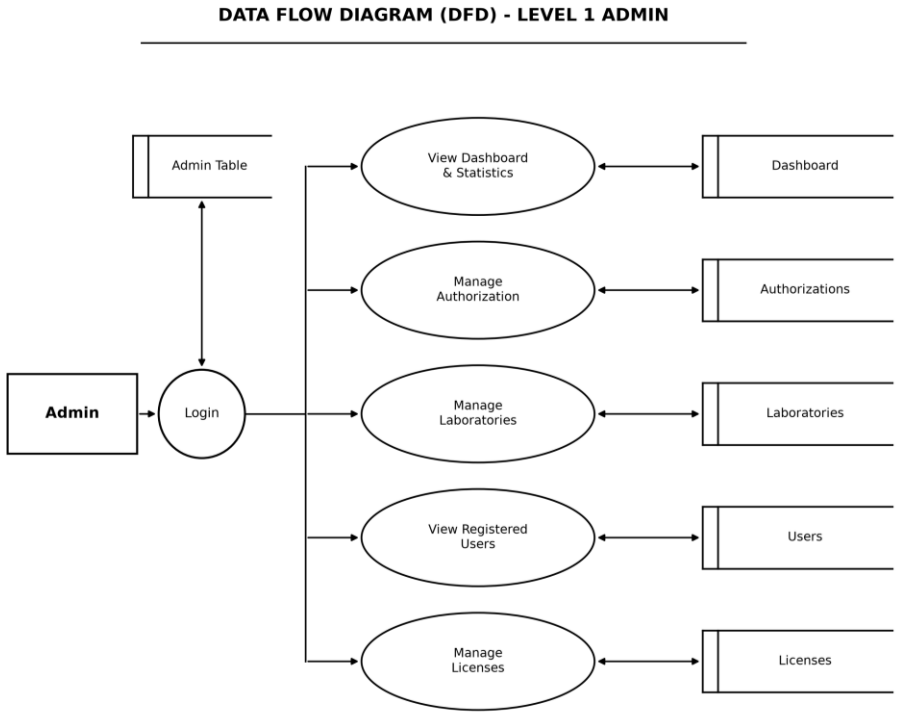
- Admin can Login and Manage Profile Information.
- Admin can Create and Manage Sub-Admin Accounts.
- Admin can Register and Manage Official Laboratory Licenses.
- Admin can Verify, Approve, Reject, or Suspend Laboratory Registrations.
- Admin can Review Attached Documents and Issue Chemical Purchase Permits.
- Admin can Manage and Update Citizen and Laboratory Accounts.
- Admin can Monitor Nationwide Purchase Transactions in Real Time.
- Admin can View Central Analytics Dashboard and Stock Statistics.
- Admin can Generate and Export Global Compliance Audit Reports.
- Admin can Logout.

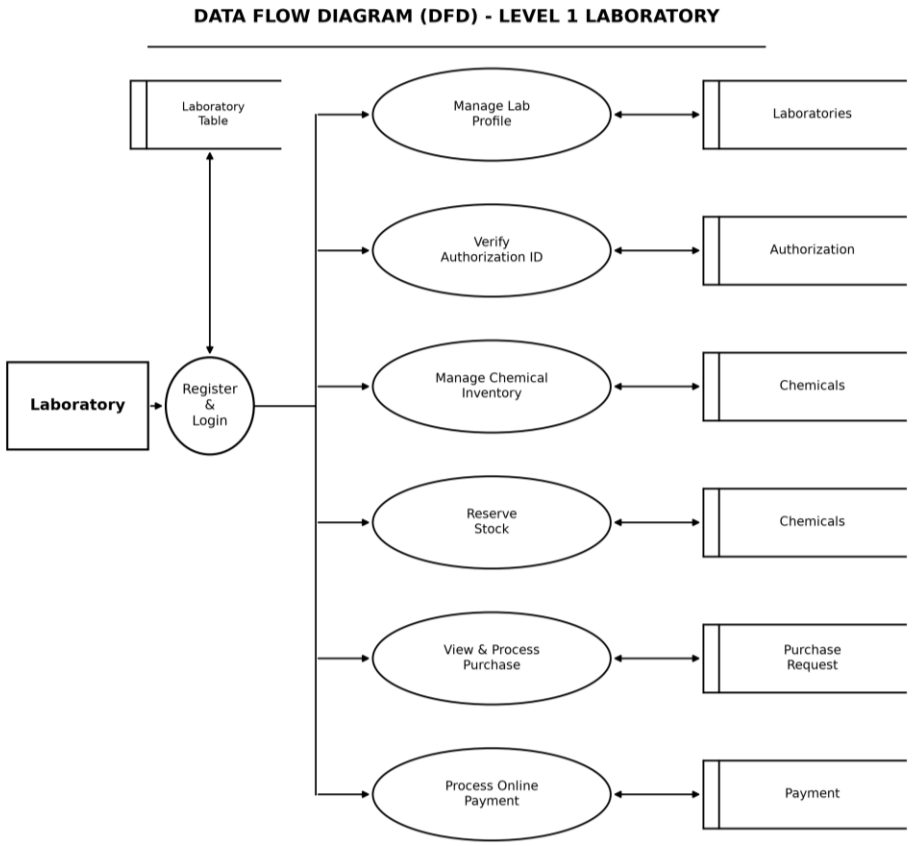
6.6 Data Flow Diagram

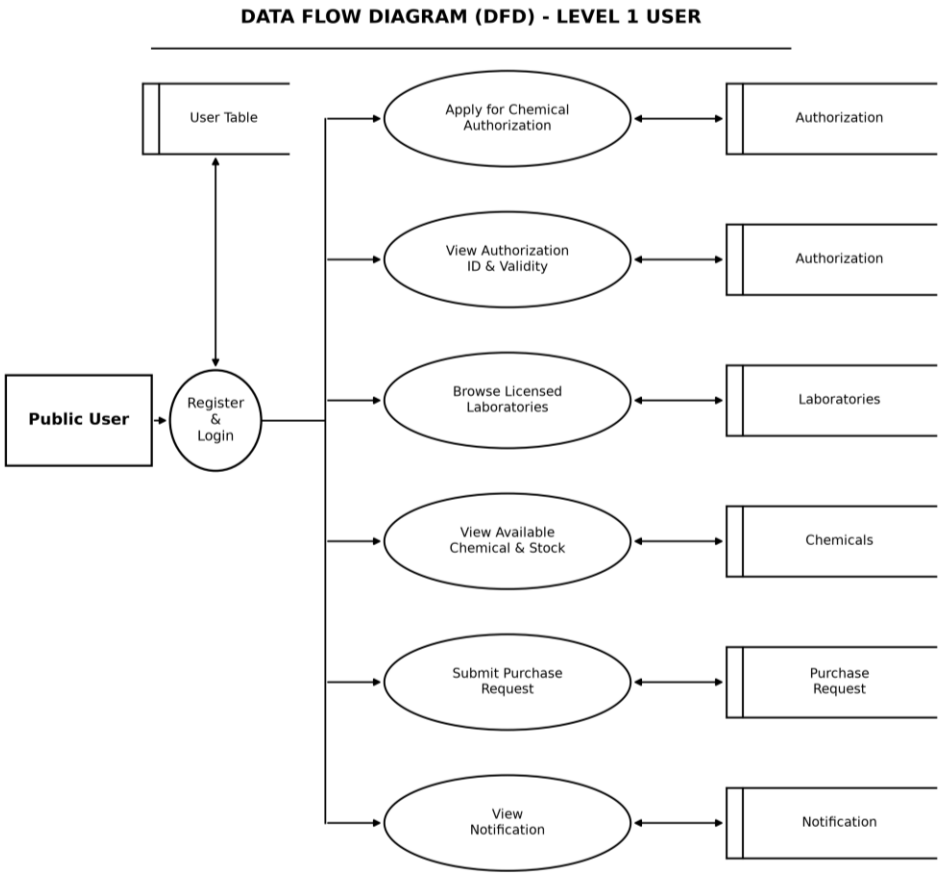
Level 0



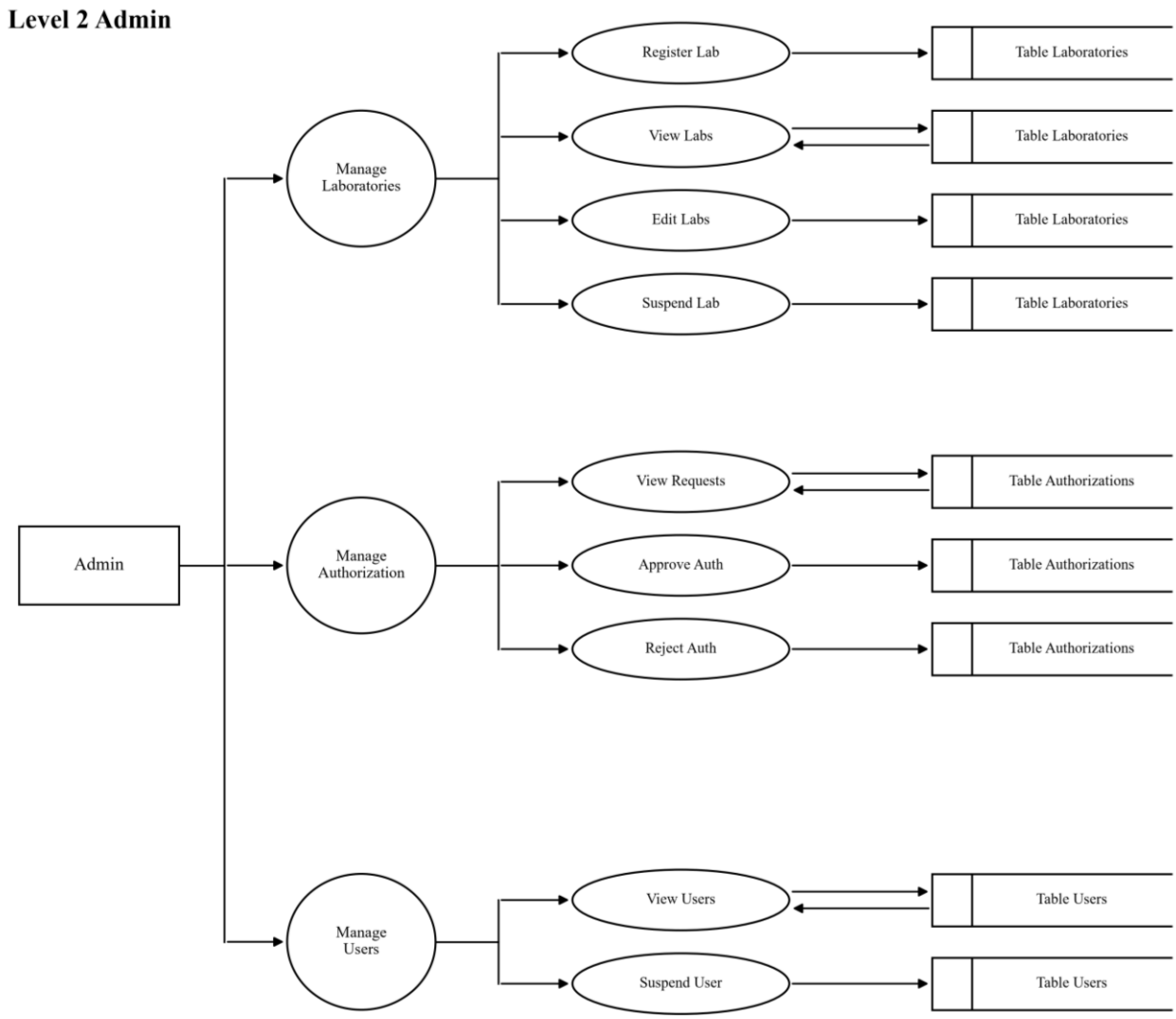
Level 1 (Admin, Laboratory, User)






DATA FLOW DIAGRAM (DFD) - LEVEL 1 USER
User Table
Public User
Register & Login
Apply for Chemical Authorization
View Authorization ID & Validity
Browse Licensed Laboratories
View Available Chemical & Stock
Submit Purchase Request
View Notification
Authorization
Authorization
Laboratories
Chemicals
Purchase Request
Notification

Level 2 (Admin, Laboratory)



Level 2 Laboratory

